

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad \sqrt{27} =$$

$$(2) \quad \sqrt{9} =$$

$$(3) \quad 5\sqrt{2} + 2\sqrt{2} =$$

$$(4) \quad 5\sqrt{7} + 3\sqrt{7} =$$

$$(5) \quad 3\sqrt{5} + 3\sqrt{5} =$$

$$(6) \quad \sqrt{8} =$$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad 3\sqrt{2} + 1\sqrt{2} =$$

$$(2) \quad \sqrt{8} =$$

$$(3) \quad 2\sqrt{6} + 1\sqrt{6} =$$

$$(4) \quad \sqrt{18} =$$

$$(5) \quad \sqrt{20} =$$

$$(6) \quad \sqrt{45} =$$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad 5\sqrt{7} + 2\sqrt{7} =$$

$$(2) \quad 2\sqrt{2} + 2\sqrt{2} =$$

$$(3) \quad 4\sqrt{2} + 3\sqrt{2} =$$

$$(4) \quad \sqrt{16} =$$

$$(5) \quad 2\sqrt{2} + 1\sqrt{2} =$$

$$(6) \quad \sqrt{50} =$$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad 3\sqrt{3} + 1\sqrt{3} =$$

$$(2) \quad \sqrt{25} =$$

$$(3) \quad \sqrt{12} =$$

$$(4) \quad \sqrt{16} =$$

$$(5) \quad \sqrt{16} =$$

$$(6) \quad \sqrt{50} =$$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad 5\sqrt{3} + 3\sqrt{3} =$$

$$(2) \quad \sqrt{16} =$$

$$(3) \quad 3\sqrt{7} + 1\sqrt{7} =$$

$$(4) \quad \sqrt{36} =$$

$$(5) \quad 3\sqrt{5} + 2\sqrt{5} =$$

$$(6) \quad \sqrt{50} =$$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad \sqrt{100} =$$

$$(2) \quad \sqrt{50} =$$

$$(3) \quad \sqrt{4} =$$

$$(4) \quad \sqrt{36} =$$

$$(5) \quad 2\sqrt{6} + 1\sqrt{6} =$$

$$(6) \quad \sqrt{9} =$$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad 4\sqrt{3} + 1\sqrt{3} =$$

$$(2) \quad \sqrt{25} =$$

$$(3) \quad 4\sqrt{3} + 3\sqrt{3} =$$

$$(4) \quad 4\sqrt{7} + 2\sqrt{7} =$$

$$(5) \quad \sqrt{20} =$$

$$(6) \quad 5\sqrt{3} + 1\sqrt{3} =$$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad \sqrt{50} =$$

$$(2) \quad 3\sqrt{2} + 1\sqrt{2} =$$

$$(3) \quad 5\sqrt{2} + 3\sqrt{2} =$$

$$(4) \quad \sqrt{20} =$$

$$(5) \quad 3\sqrt{2} + 1\sqrt{2} =$$

$$(6) \quad \sqrt{100} =$$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad 5\sqrt{6} + 2\sqrt{6} =$$

$$(2) \quad \sqrt{49} =$$

$$(3) \quad \sqrt{50} =$$

$$(4) \quad \sqrt{45} =$$

$$(5) \quad \sqrt{8} =$$

$$(6) \quad 2\sqrt{5} + 3\sqrt{5} =$$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad \sqrt{100} =$$

$$(2) \quad 2\sqrt{3} + 2\sqrt{3} =$$

$$(3) \quad \sqrt{18} =$$

$$(4) \quad 3\sqrt{5} + 1\sqrt{5} =$$

$$(5) \quad 5\sqrt{5} + 2\sqrt{5} =$$

$$(6) \quad \sqrt{32} =$$

Section 6. Square Roots

Simplify the square root expressions.

(1) $\sqrt{12} =$

(2) $\sqrt{25} =$

(3) $\sqrt{64} =$

(4) $\sqrt{4} =$

(5) $\sqrt{64} =$

(6) $\sqrt{12} =$

Section 6. Square Roots

Simplify the square root expressions.

(1) $3\sqrt{5} + 3\sqrt{5} =$

(2) $\sqrt{25} =$

(3) $\sqrt{36} =$

(4) $\sqrt{45} =$

(5) $\sqrt{81} =$

(6) $\sqrt{45} =$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad 3\sqrt{2} + 3\sqrt{2} =$$

$$(2) \quad \sqrt{49} =$$

$$(3) \quad 2\sqrt{2} + 2\sqrt{2} =$$

$$(4) \quad \sqrt{8} =$$

$$(5) \quad 4\sqrt{5} + 1\sqrt{5} =$$

$$(6) \quad \sqrt{9} =$$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad \sqrt{25} =$$

$$(2) \quad 2\sqrt{6} + 1\sqrt{6} =$$

$$(3) \quad \sqrt{36} =$$

$$(4) \quad 4\sqrt{6} + 1\sqrt{6} =$$

$$(5) \quad 3\sqrt{3} + 2\sqrt{3} =$$

$$(6) \quad \sqrt{16} =$$

Section 6. Square Roots

Simplify the square root expressions.

(1) $\sqrt{12} =$

(2) $5\sqrt{6} + 2\sqrt{6} =$

(3) $5\sqrt{3} + 3\sqrt{3} =$

(4) $\sqrt{49} =$

(5) $3\sqrt{3} + 2\sqrt{3} =$

(6) $5\sqrt{5} + 2\sqrt{5} =$

Section 6. Square Roots

Simplify the square root expressions.

(1) $\sqrt{8} =$

(2) $3\sqrt{2} + 3\sqrt{2} =$

(3) $2\sqrt{6} + 1\sqrt{6} =$

(4) $\sqrt{25} =$

(5) $4\sqrt{5} + 1\sqrt{5} =$

(6) $\sqrt{100} =$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad 3\sqrt{6} + 1\sqrt{6} =$$

$$(2) \quad \sqrt{32} =$$

$$(3) \quad \sqrt{36} =$$

$$(4) \quad \sqrt{81} =$$

$$(5) \quad \sqrt{81} =$$

$$(6) \quad \sqrt{81} =$$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad \sqrt{25} =$$

$$(2) \quad \sqrt{81} =$$

$$(3) \quad 5\sqrt{5} + 2\sqrt{5} =$$

$$(4) \quad 4\sqrt{7} + 3\sqrt{7} =$$

$$(5) \quad \sqrt{20} =$$

$$(6) \quad \sqrt{18} =$$

Section 6. Square Roots

Simplify the square root expressions.

(1) $\sqrt{25} =$

(2) $\sqrt{81} =$

(3) $\sqrt{27} =$

(4) $\sqrt{45} =$

(5) $\sqrt{32} =$

(6) $\sqrt{8} =$

Section 6. Square Roots

Simplify the square root expressions.

(1) $\sqrt{4} =$

(2) $\sqrt{49} =$

(3) $\sqrt{9} =$

(4) $\sqrt{32} =$

(5) $3\sqrt{6} + 3\sqrt{6} =$

(6) $\sqrt{27} =$

Section 6 - Solutions

51a-1: $3\sqrt{3}$

51a-2: 3

51a-3: $7\sqrt{2}$

51a-4: $8\sqrt{7}$

51a-5: $6\sqrt{5}$

51a-6: $2\sqrt{2}$

51b-1: $4\sqrt{3}$

51b-2: 5

51b-3: $2\sqrt{3}$

51b-4: 4

51b-5: 4

51b-6: $5\sqrt{2}$

52a-1: $4\sqrt{2}$

52a-2: $2\sqrt{2}$

52a-3: $3\sqrt{6}$

Section 6 - Solutions (continued)

53b-1: $5\sqrt{2}$

53b-2: $4\sqrt{2}$

53b-3: $8\sqrt{2}$

53b-4: $2\sqrt{5}$

53b-5: $4\sqrt{2}$

53b-6: 10

54a-1: 10

54a-2: $5\sqrt{2}$

54a-3: 2

54a-4: 6

54a-5: $3\sqrt{6}$

54a-6: 3

54b-1: $5\sqrt{3}$

54b-2: 5

54b-3: $7\sqrt{3}$

Section 6 - Solutions (continued)

54b-4: $6\sqrt{7}$

54b-5: $2\sqrt{5}$

54b-6: $6\sqrt{3}$

55a-1: $7\sqrt{6}$

55a-2: 7

55a-3: $5\sqrt{2}$

55a-4: $3\sqrt{5}$

55a-5: $2\sqrt{2}$

55a-6: $5\sqrt{5}$

55b-1: $6\sqrt{5}$

55b-2: 5

55b-3: 6

55b-4: $3\sqrt{5}$

55b-5: 9

55b-6: $3\sqrt{5}$

Section 6 - Solutions (continued)

52a-4: $3\sqrt{2}$

52a-5: $2\sqrt{5}$

52a-6: $3\sqrt{5}$

52b-1: $7\sqrt{7}$

52b-2: $4\sqrt{2}$

52b-3: $7\sqrt{2}$

52b-4: 4

52b-5: $3\sqrt{2}$

52b-6: $5\sqrt{2}$

53a-1: $8\sqrt{3}$

53a-2: 4

53a-3: $4\sqrt{7}$

53a-4: 6

53a-5: $5\sqrt{5}$

53a-6: $5\sqrt{2}$

Section 6 - Solutions (continued)

56a-1:	10
56a-2:	$4\sqrt{3}$
56a-3:	$3\sqrt{2}$
56a-4:	$4\sqrt{5}$
56a-5:	$7\sqrt{5}$
56a-6:	$4\sqrt{2}$
56b-1:	$2\sqrt{3}$
56b-2:	5
56b-3:	8
56b-4:	2
56b-5:	8
56b-6:	$2\sqrt{3}$
57a-1:	$6\sqrt{2}$
57a-2:	7
57a-3:	$4\sqrt{2}$

Section 6 - Solutions (continued)

58b-1:	$2\sqrt{3}$
58b-2:	$7\sqrt{6}$
58b-3:	$8\sqrt{3}$
58b-4:	7
58b-5:	$5\sqrt{3}$
58b-6:	$7\sqrt{5}$
59a-1:	$4\sqrt{6}$
59a-2:	$4\sqrt{2}$
59a-3:	6
59a-4:	9
59a-5:	9
59a-6:	9
59b-1:	2
59b-2:	7
59b-3:	3

Section 6 - Solutions (continued)

59b-4: $4\sqrt{2}$

59b-5: $6\sqrt{6}$

59b-6: $3\sqrt{3}$

60a-1: 5

60a-2: 9

60a-3: $7\sqrt{5}$

60a-4: $7\sqrt{7}$

60a-5: $2\sqrt{5}$

60a-6: $3\sqrt{2}$

60b-1: 5

60b-2: 9

60b-3: $3\sqrt{3}$

60b-4: $3\sqrt{5}$

60b-5: $4\sqrt{2}$

60b-6: $2\sqrt{2}$

Section 6 - Solutions (continued)

57a-4: $2\sqrt{2}$

57a-5: $5\sqrt{5}$

57a-6: 3

57b-1: $2\sqrt{2}$

57b-2: $6\sqrt{2}$

57b-3: $3\sqrt{6}$

57b-4: 5

57b-5: $5\sqrt{5}$

57b-6: 10

58a-1: 5

58a-2: $3\sqrt{6}$

58a-3: 6

58a-4: $5\sqrt{6}$

58a-5: $5\sqrt{3}$

58a-6: 4